Notes: Einav, Finkelstein, Ryan, Schrimpf, and Cullen (AER, 2013)

Selection on Moral Hazard in Health Insurance

Contents

1 Big picture

- **Question.** The paper asks whether people choose health insurance coverage partly because they know how much their own medical spending will respond to insurance coverage. This is the phenomenon the authors call *selection on moral hazard*.
- **Key distinction.** Standard adverse selection emphasizes the *level* of expected medical spending: sicker people buy more insurance. This paper adds selection on the *slope*: people with a larger behavioral spending response to insurance may also prefer more generous coverage.
- **Setting.** The empirical setting is employer-provided health insurance at Alcoa. The authors observe plan menus, plan choices, demographics, health risk scores, and claim-level medical spending for employees and covered dependents.
- **Source of variation.** Alcoa introduced a new set of PPO options with higher cost sharing. Because benefits changed at different times across unions, the staggered transition provides quasi-experimental variation in insurance generosity.
- **Model primitives.** Individuals differ in expected health risk, risk aversion, and moral hazard. In the model, health risk is a *level* parameter, while moral hazard is a *slope* parameter measuring the incremental spending induced by more generous insurance.
- **Descriptive result.** Moving workers to the new, higher-cost-sharing options reduces medical spending. The difference-in-differences estimates imply a sizable average moral hazard effect.
- **Structural result.** Moral hazard is highly heterogeneous, and people with larger estimated behavioral responses are more likely to choose more comprehensive coverage.
- **Policy implication.** If high-deductible plans are introduced, the people who voluntarily choose them are not random. They tend to have lower moral hazard responses, so the spending reduction from plan choice can be smaller than what one would infer from an average moral hazard estimate.
- **Economic takeaway.** Selection and moral hazard are not independent problems. The same private information that affects insurance choice can include information about how strongly the person will respond to insurance.

2 Incremental contribution of the paper

- **From separate frictions to an integrated friction.** Much health-insurance work studies adverse selection and moral hazard separately. This paper shows that the two can interact because people may select coverage based on their expected moral hazard response.
- **A levels-versus-slopes contribution.** The paper decomposes adverse selection into selection on expected health spending levels and selection on the slope of spending with respect to insurance coverage.
- **A treatment-effect interpretation.** Selection on moral hazard is an insurance-market version of selection on heterogeneous treatment effects. Individuals may select into treatment, here more generous insurance, partly because their individual treatment effect is large.

- **A data contribution.** The paper combines employee-level insurance choices, claims, plan menus, and a plausibly exogenous staggered change in cost sharing at a large employer.
- **A structural measurement contribution.** The model recovers a joint distribution of expected health risk, risk aversion, and moral hazard. This lets the authors quantify not only average moral hazard but also heterogeneity and selection on that heterogeneity.
- **A counterfactual contribution.** The paper studies what happens when a high-deductible plan is introduced. The results show why average moral hazard estimates from randomized or quasi-random cost-sharing changes may overstate the effects of voluntary plan choice.
- **A welfare and contract-design contribution.** The analysis clarifies that monitoring technologies usually viewed as moral-hazard tools can also affect adverse selection by reducing selection on the moral hazard dimension.
- **A methodological contribution.** Instead of adding an i.i.d. plan-choice error to make choices easy to fit, the authors rely on rich unobserved heterogeneity in economically interpretable primitives.

3 Data and measurement

3.1 Alcoa employee panel

- The main data cover U.S.-based Alcoa workers and their covered dependents.
- The baseline structural analysis uses 2003 and 2004 data: 7,570 employee-years and 4,477 unique employees.
- The descriptive difference-in-differences analysis is also shown for a broader 2003–2006 sample to improve precision.
- The baseline sample focuses on roughly 4,000 unionized workers each year. Their benefit changes were tied to union contract timing.
- The data include plan choices, plan menus, claims, annual medical spending, demographics, union affiliation, earnings, tenure, coverage tier, and health risk scores.

Interpretation: The central empirical advantage is that plan menus change for institutional reasons, while individual choices and spending are observed at high frequency and with detailed claims data.

3.2 Health risk and spending measures

- The main spending outcome is total annual medical spending for the employee plus any covered dependents.
- Spending is highly skewed. Many employees spend little or nothing, while a small upper tail accounts for a large share of spending.
- The health risk score is a predicted-spending measure constructed from prior diagnoses, claims, age, and gender.

- The health risk score is important because it captures expected medical need before the coverage choice is analyzed.

Interpretation: The paper needs to distinguish high spending because someone is sick from high spending because someone responds strongly to insurance. The health risk score helps separate the level of risk from the behavioral response.

3.3 Old and new insurance options

- Under the old benefits, workers chose among three PPO options.
- Under the new benefits, workers chose among five PPO options.
- The plans are PPO plans that differ mainly in cost sharing: deductibles, out-of-pocket maxima, and premiums.
- The new options generally increase consumer cost sharing. The most important move is toward higher deductibles and higher out-of-pocket exposure.
- The authors take coverage tier as given, because single, employee plus spouse, employee plus children, and family coverage are primarily driven by household structure.

Interpretation: The menu change creates a useful empirical setting: plan generosity changes, but the provider network and other non-cost-sharing features are broadly comparable within the PPO setting.

3.4 Plan choice and outcomes

- The key choice outcome is which health plan the employee selects from the available menu.
- The key spending outcome is annual medical spending after plan choice.
- The model jointly explains plan choice and spending.
- The plan-choice data identify willingness to pay for coverage, while the spending data identify how medical utilization responds to coverage.

Interpretation: The paper's main contribution requires both parts of the data. Spending variation alone can identify moral hazard; plan-choice variation is needed to ask whether people select on that moral hazard.

4 Models and empirical specifications

4.1 Period-2 utilization model

The individual has realized health λ, moral hazard type ω, and insurance coverage j. The second-period utility from medical utilization m is

$$u(m;\lambda,\omega,j) = \left[(m-\lambda) - \frac{1}{2\omega}(m-\lambda)^2\right] + \left[y - c_j(m) - p_j\right].$$

The utilization choice is

$$m^*(\lambda, \omega, j) = \arg\max_{m \geq 0} \nu(m; \lambda, \omega, j).$$

For intuition, with a linear contract $c_j(m) = c \cdot m$, the first-order condition gives

$$m^*(\lambda, \omega, c) = \max[0, \lambda + \omega(1 - c)].$$

- λ is the level of health care need. It can be interpreted as nondiscretionary spending.
- ω is the behavioral response to insurance. Higher ω means a larger spending increase when out-of-pocket prices fall.
- Full insurance corresponds to $c = 0$; no insurance corresponds to $c = 1$.
- Ignoring the zero-spending truncation, spending is λ under no insurance and $\lambda + \omega$ under full insurance.

Interpretation: This is the core levels-versus-slopes model. λ says how expensive the person is without insurance; ω says how much more the person spends when insurance becomes generous.

4.2 Period-1 coverage choice

At the time of plan choice, the individual does not know the exact future realization of λ. The individual has beliefs $F_\lambda(\cdot)$, moral hazard type ω, and CARA risk aversion ψ. Expected utility from plan j is

$$v_j(F_\lambda, \omega, \psi) = -\int \exp\left(-\psi u^*(\lambda, \omega, j)\right) dF_\lambda(\lambda),$$

where

$$u^*(\lambda, \omega, j) = \nu(m^*(\lambda, \omega, j); \lambda, \omega, j).$$

The chosen plan is

$$j^*(F_\lambda, \omega, \psi) = \arg\max_{j \in J} v_j(F_\lambda, \omega, \psi).$$

- Higher expected health risk generally increases demand for generous coverage.
- Higher risk aversion generally increases demand for generous coverage.
- Higher moral hazard also increases willingness to pay for coverage because the person expects to consume more care when coverage is generous.

Interpretation: The last comparative static is the paper's focus. If higher ω people value generous insurance more, then plan choice can reveal selection on moral hazard.

4.3 Econometric heterogeneity

The structural model lets workers differ across several latent dimensions.

- Health risk λ_{it} is drawn from a shifted lognormal distribution:

$$\log(\lambda_{it} - \kappa_{\lambda,i}) \sim N(\mu_{\lambda,it}, \sigma^2_{\lambda,i}).$$

- Expected health risk is
$$\bar{\lambda}(\mu_\lambda, \sigma_\lambda, \kappa_\lambda) = \exp\left(\mu_\lambda + \frac{1}{2}\sigma_\lambda^2\right) + \kappa_\lambda.$$
- Moral hazard ω_i and risk aversion ψ_i vary across individuals.
- The average health-risk parameter, log moral hazard, and log risk aversion are modeled as jointly normal conditional on observables.
- The model allows correlations among expected health, moral hazard, and risk aversion.

Interpretation: The flexible correlation structure is essential. Whether high-moral-hazard people choose more insurance depends not only on the direct effect of ω, but also on how ω is correlated with health risk and risk aversion.

4.4 Estimation method

- The model is estimated by Markov Chain Monte Carlo Gibbs sampling.
- The latent variables include individual health realizations, health expectation parameters, moral hazard, and risk aversion.
- Conditional on the latent variables, parameter sampling is standard.
- Conditional on the parameters, latent variables are sampled to rationalize observed plan choices and spending.
- The authors avoid a plan-specific i.i.d. error term because the plan options are financially rankable and differ primarily in cost sharing.

Interpretation: The estimator is not just fitting plan shares. It recovers economically meaningful unobserved heterogeneity that jointly explains choices and subsequent medical spending.

5 Identification and empirical design

5.1 Average moral hazard

The paper first uses a difference-in-differences design. Let treatment group denote workers whose union contract moves them to the new options. A simple estimating equation can be written as

$$Spending_{it} = \alpha + \delta NewOptions_{it} + \gamma_t + \eta_g + \varepsilon_{it},$$

where γ_t are year effects and η_g are treatment-group effects.

- The coefficient δ measures the spending change associated with the new, higher-cost-sharing menu.
- The staggered timing comes from union contract expiration dates.
- The 2003–2004 sample gives the clean baseline but is imprecise because relatively few workers switch in 2004.

- The 2003–2006 sample provides more precision because more workers transition to the new options.

Interpretation: This descriptive design identifies average moral hazard from a menu shift. It does not, by itself, identify individual heterogeneity in moral hazard or selection on that heterogeneity.

5.2 Selection on moral hazard

- Selection on health risk means people with high expected λ choose more generous coverage.
- Selection on risk aversion means people with high ψ choose more generous coverage.
- Selection on moral hazard means people with high ω choose more generous coverage.
- The empirical problem is that plan choice is endogenous and health spending is noisy.
- The structural model uses both plan-choice and spending data to separate λ, ψ, and ω.

Interpretation: A high spender in a generous plan may be sick, risk averse, responsive to coverage, or some combination of all three. The model is designed to separate these channels.

5.3 Ideal identification intuition

With ideal data, suppose each worker is observed for a long time before and after an exogenous coverage change.

- Before-and-after spending distributions would identify the worker's health distribution and moral hazard parameter.
- The change in spending around a coverage change identifies ω_i.
- Conditional on health risk and moral hazard, plan choice identifies risk aversion ψ_i.
- With a continuous menu of plans, individual risk aversion would be more directly identified.

Interpretation: The actual data are less ideal: there are only two years in the baseline structural sample and plan menus are discrete. This is why the paper imposes parametric structure and estimates the joint distribution.

5.4 Important assumptions and limitations

- The model assumes individuals have correct beliefs about their health risk distribution.
- The quantitative results are specific to Alcoa, the time period, the plan designs, and the unionized worker sample.
- The model abstracts from richer dynamic features of spending during the year.
- The paper does not claim that all medical spending responses are wasteful in a normative sense.
- The structural estimates depend on the functional-form assumptions used to separate latent health risk, risk aversion, and moral hazard.

Interpretation: The paper is strongest as evidence that selection on moral hazard exists and can be quantitatively important in this setting. Its exact magnitudes should not be automatically generalized to all health-insurance markets.

6 Tables and figures

6.1 Table 1 and Figure 1: Sample composition and spending distribution

Read:

- Table 1 summarizes the 2003 baseline sample and the treatment groups based on when workers switch to the new options.
- The average worker is about 41 years old, mostly male, and has about ten years of tenure at Alcoa.
- The treatment groups differ in observables, which motivates using treatment-group fixed effects and control variables.
- Figure 1 shows that medical spending is extremely skewed for both single and nonsingle coverage.
- Nonsingle coverage has much higher average and median spending than single coverage.

Economic message: The empirical setting has rich variation but also strong skewness in spending. Mean spending is heavily influenced by the upper tail, so a model of spending needs to handle zero and high-spending observations.

TABLE 1—SUMMARY STATISTICS FOR 2003 SAMPLE

	Baseline sample	Switched: In 2004	Switched: In 2005	Switched: In 2006	Switched: After 2006
	(1)	(2)	(3)	(4)	(5)
Observations	3,995	682	974	1,075	1,264
Average age	41.3	44.5	39.7	38.3	43.3
Average annual income	31,292	39,715	25,532	29,952	32,316
Average tenure with Alcoa	10.2	15.5	8.2	5.7	12.7
Fraction male	0.84	0.96	0.73	0.86	0.85
Fraction white	0.72	0.85	0.44	0.82	0.79
Fraction single coverage	0.23	0.21	0.25	0.23	0.22
Average health risk score[a]	0.95	1.06	0.91	0.86	1.01
Avg. number of insured family members (if > 1)	2.8	2.7	2.8	2.9	2.6
Total annual medical spending (US$)[b]	5,283	5,194	5,364	5,927	4,717

Notes: Column 1 presents statistics based on the 2003 data for our baseline sample, which covers all hourly union workers not covered by the Master Steelworker's Agreement (except those that get dropped in the process of the data cleaning described in the text). Columns 2–5 partition our baseline sample based on the year in which employees were switched to the new set of health insurance options.

[a] Health risk score is normalized, so that 1 indicates the expected medical expenditure for a random draw from a nationally representative under-65 population. To construct this table, we assign each employee the average risk score of all covered family members.

[b] Total annual medical spending is for employees and any covered dependents.

(a) Table 1: Summary statistics for the 2003 sample.

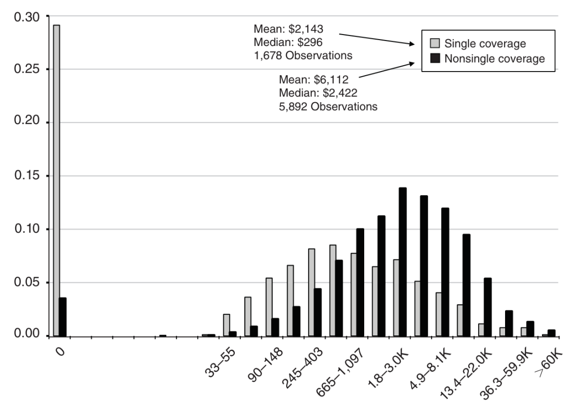

FIGURE 1. THE CROSS-SECTIONAL DISTRIBUTION OF MEDICAL EXPENDITURE

Notes: The figure presents the distribution of total annual medical expenditure for each employee (and any covered dependents) in our baseline sample. The graph uses a log scale, such that the second bin covers expenditure lower than exp(0.5), the next covers expenditures between exp(0.5) and exp(1), and so on; the *x*-axis labels show the corresponding dollar amounts of selected bins. An observation is an employee-year, pooling data from 2003 and 2004. The gray bars correspond to employees with a single coverage, while the black bars correspond to employees who also covered additional dependents (spouse, children, or both).

(b) Figure 1: Cross-sectional distribution of medical expenditure.

6.2 Table 2: Old and new health plans

Read:

- Table 2 describes the old and new PPO plan menus for single and nonsingle coverage.
- The old menu had three options; the new menu has five options.
- The new options introduce higher deductibles and generally higher consumer cost sharing.
- Under the new options, many employees still choose the highest coverage option, but the menu gives them meaningful opportunities to trade off premiums and cost sharing.
- The table also calculates the average share of spending paid out of pocket under each plan, holding realized claims fixed.

Economic message: Table 2 is the contract-design foundation of the paper. The move to the new menu changes the price of care faced by workers and creates the variation needed to identify behavioral spending responses.

TABLE 2—OLD AND NEW HEALTH PLANS

	Original plan options			New plan options				
	Option 1	Option 2	Option 3	Option 1[a]	Option 2	Option 3	Option 4	Option 5
Panel A. Single coverage ($N = 1{,}679$)								
Plan features:								
Deductible	1,000	0	0	1,500	750	500	250	0
Out-of-pocket maximum	5,000	2,500	1,000	4,500	3,750	3,500	2,750	2,500
Average share of spending paid out of pocket[b]	0.580	0.150	0.111	0.819	0.724	0.660	0.535	0.112
Employee premium[c]	0	351	1,222	0	132	224	336	496
Fraction choosing each option[d]	3.3%	63.5%	33.2%	14.1%	0.0%	2.2%	37.8%	45.9%
Panel B. Nonsingle coverage ($N = 5{,}895$)								
Plan features:								
Deductible	2,000	0	0	3,000	1,500	1,000	500	0
Out-of-pocket maximum	10,000	5,000	2,000	9,000	7,500	7,000	5,500	5,000
Average share of spending paid out of pocket[b]	0.495	0.130	0.098	0.732	0.600	0.520	0.387	0.111
Employee premium[c]	0	354	1,297	0	364	620	914	1,306
Fraction choosing each option[d]	0.6%	56.1%	43.3%	3.9%	0.6%	1.8%	24.4%	69.3%

Notes: The table summarizes the key features of the original and new health insurance coverage options. The features shown apply to in-network spending. Not shown are coinsurance rates (applied to those who reached the deductible but have yet to reach the out-of-pocket maximum), which are 10 percent in all plans (old and new). There are some other small differences between the original and new options that are associated with out-of-network spending, preventive care, and certain treatments associated with copays rather than coinsurance in the original set of options. See text for further details.

[a] The New Option 1 includes a health reimbursement arrangement (HRA). Every year the employer sets aside \$750 (for single; \$1,250 for nonsingle coverage) that the employee can use (tax free) to pay for a variety of expenses such as deductibles and coinsurance payments. Unused HRA funds roll over to future years and, eventually, can be used during retirement to finance health insurance, provided through the company or through COBRA. Our baseline model abstracts from the HRA component of New Option 1.

[b] To compute the average share of spending out of pocket, we use the 2003 claims from all individuals in the baseline sample and apply each option's coverage details to this (common) sample. We then compute, for each option, the ratio of the resultant out-of-pocket expenses to the total claim mounts, and report the average (across employees in 2003) for each option. As a result, our computed average share of spending out of pocket abstracts from any differential behavioral effect of each contract. It does, however, account for the unmodeled small differences between the new and original options described above and in the text.

[c] Premiums are normalized so that the lowest coverage is free for all employees. This is true in both the original and new options, up to small variation of several hundred dollars across employees. We report the average premium for employees in the baseline sample, pooling 2003 and 2004. Premiums vary by coverage tier; there is also some additional variation (across employees within coverage tier) in the incremental premiums associated with greater coverage options. The variation is based on the business unit to which each employees belongs (see Einav, Finkelstein, and Cullen 2010).

[d] Statistics are based on all employee in the baseline sample, pooling 2003 and 2004.

Table 2: Old and new health plans.

6.3 Table 3 and Table 4: Plan transitions and positive correlation evidence

Read:

- Table 3 shows plan transitions. Workers who stay in the old options mostly keep the same relative coverage level.
- Workers moved to the new options do not simply map mechanically into the same rank. This suggests active plan choice under the new menu.

- Table 4 shows that people choosing higher coverage have higher contemporaneous medical spending.
- This positive coverage-spending correlation is consistent with adverse selection, moral hazard, or both.

Economic message: Table 4 establishes asymmetric-information-type patterns, but it does not separate selection from moral hazard. The rest of the paper's design is needed to decompose the positive correlation.

TABLE 3—PLAN TRANSITIONS (*Percent*)

	Old options in 2004	
	Highest coverage	All other coverages
2003		
Highest coverage	40.0	0.5
All other coverages	0.6	58.9
	New options in 2004	
	Highest coverage	All other coverages
2003		
Highest coverage	32.0	15.8
All other coverages	27.8	24.5

Notes: The table shows transition matrices across plan options for those in the old options in both 2003 and 2004 (top panel) and those who are switched to the new options in 2004 (bottom panel). Under the original options, the highest coverage is option 3. Under the new options, the highest coverage in option 5. See Table 2 for coverage details. The sample is limited to the 6,186 employees (82 percent of the baseline sample) who are in the data in both 2003 and 2004.

(a) Table 3: Plan transitions.

TABLE 4—SPENDING PATTERNS BY COVERAGE LEVEL

	Single coverage			Nonsingle coverage		
	Count	Mean	Median	Count	Mean	Median
Original plan options						
Highest coverage	512	3,130	557	2,318	6,634	2,670
All other coverages	1,031	1,795	233	3,035	5,768	2,288
New plan options						
Highest coverage	62	1,650	447	375	6,858	2,630
All other coverages	73	560	52	164	3,405	1,481

Notes: The table shows (contemporaneous) spending by coverage choice. Under the original options, the highest coverage is option 3. Under the new options, the highest coverage is option 5. See Table 2 for coverage details.

(b) Table 4: Spending patterns by coverage level.

6.4 Table 5 and Table 6: Difference-in-differences evidence

Read:

- Table 5 shows raw difference-in-differences patterns for the 2003–2004 baseline sample.
- Spending falls for workers switched to the new options and is roughly flat or slightly rising for workers who remain under the old options.
- The decline is visible across many parts of the spending distribution, not only at the mean.
- Table 6 reports regression estimates of the spending reduction from moving to the new options.
- The 2003–2006 estimates are more precise and suggest economically meaningful reductions in spending.

Economic message: These tables provide the paper's reduced-form moral hazard evidence. Higher cost sharing lowers medical spending, which motivates measuring who has larger or smaller behavioral responses.

TABLE 5—BASIC DIFFERENCE-IN-DIFFERENCES IN BASELINE SAMPLE

	Observations	Mean	Fraction with zero spending	10th	25th	50th	75th	90th
Control (switched after 2004)								
2003 spending	3,313	5,300	0.09	52	426	1,775	5,178	11,984
2004 spending	2,901	5,250	0.09	55	517	1,889	5,589	12,253
Treated (switched in 2004)								
2003 spending	682	5,201	0.08	79	581	1,957	5,048	12,644
2004 spending	674	4,856	0.10	22	447	1,610	4,622	9,467
Treated-Control differences (levels)								
2003 spending		−99	−0.01	27	155	182	−130	660
2004 spending		−394	0.01	−33	−70	−279	−967	−2,786
2004–2003 Difference (levels)								
Control		−50	0.00	3	91	114	411	269
Treated		−345	0.02	−57	−134	−347	−426	−3,177
Difference-in-differences (levels)		−295	0.02	−60	−225	−461	−837	−3,446
Difference (percentages)								
Control		−0.9%	0.0%	5.8%	21.4%	6.4%	7.9%	2.2%
Treated		−6.6%	22.0%	−72.2%	−23.1%	−17.7%	−8.4%	−25.1%
Difference-in-differences (percentages)		−5.7%	22.0%	−77.9%	−44.4%	−24.2%	−16.4%	−27.4%

(a) Table 5: Basic difference-in-differences in the baseline sample.

TABLE 6—DIFFERENCE-IN-DIFFERENCES ESTIMATES OF IMPACT OF CHANGE IN HEALTH INSURANCE OPTIONS ON ANNUAL MEDICAL SPENDING

	2003–2004 sample			2003–2006 sample		
	OLS in levels (1)	OLS in logs (2)	QMLE-Poisson (3)	OLS in levels (4)	OLS in logs (5)	QMLE-Poisson (6)
Estimated treatment effect	−297.2	−0.35	−0.06	−591.8	−0.175	−0.114
	(753.7)	(0.19)	(0.15)	(264.2)	(0.12)	(0.048)
	[0.70]	[0.08]	[0.69]	[0.034]	[0.17]	[0.018]
Mean dependent variable	5,232	6.91	5,232	5,392	6.9	5,392
Observations	7,570	7,570	7,570	14,638	14,638	14,638

Notes: The table shows the difference-in-differences estimate of the spending reduction associated with moving from the old options to the new options. The unit of observation is an employee-year. Dependent variable is the total annual medical spending for each employee and any covered dependents (or log of 1 + total spending in column 2 and column 5). The coefficient shown is the coefficient on an indicator variable that is equal to 1 if the employee's treatment group is offered the new health insurance options that year, and 0 otherwise. All regressions include year and treatment group fixed effects. We classify employees into one of four possible treatment groups—switched in 2004, switched in 2005, switched in 2006, or switched later—based on his union affiliation, which determines the year in which he is switched to the new health insurance options. Estimation is either by OLS or QMLE Poisson as indicated in the column headings. Standard errors (in parentheses) are adjusted for an arbitrary variance-covariance matrix within each of the 28 unions; p-values are in square brackets. Columns 1–3 show estimates for the 2003–2004 sample; columns 4–6 expand the sample to include 2003–2006.

(b) Table 6: Difference-in-differences estimates of spending effects.

6.5 Table 7A and Table 7B: Structural parameter estimates

Read:

- Table 7A reports the structural parameter estimates for health risk, moral hazard, and risk aversion.
- The model allows observable variables and latent heterogeneity to shift these primitives.
- Table 7B translates the parameter estimates into more interpretable quantities.
- The average expected health risk is about \$4,340 per employee-year.
- The average moral hazard parameter is about \$1,330, and the standard deviation of moral hazard is much larger than the mean.
- The estimated correlation between expected health risk and moral hazard is positive, while the correlations of risk aversion with health risk and moral hazard are negative.

Economic message: The structural estimates show that moral hazard is not a single average number. It varies greatly across individuals, which makes selection on the moral hazard dimension quantitatively possible.

Table 7A—Parameter Estimates

	μ_λ (Health risk)	κ_λ (Health risk)	$\ln(\omega)$ (Moral hazard)	$\ln(\psi)$ (Risk aversion)
Mean shifters				
Constant	6.11	−389	5.31	−5.57
	(0.14)	(73)	(0.24)	(0.10)
Coverage tier				
Single	(Omitted)	(Omitted)	(Omitted)	(Omitted)
Family	0.19	57	−0.58	−0.88
	(0.08)	(51)	(0.18)	(0.07)
Emp+Spouse	0.27	44	−0.66	−0.95
	(0.09)	(53)	(0.22)	(0.07)
Emp+Children	0.24	185	−0.28	−0.91
	(0.08)	(47)	(0.21)	(0.06)
Treatment group				
Switch 2004	−0.01	−278	−0.24	−0.31
	(0.07)	(43)	(0.11)	(0.05)
Switch 2005	−0.10	−78	0.07	−0.23
	(0.06)	(38)	(0.12)	(0.05)
Switch 2006	0.12	−94	0.01	−0.07
	(0.07)	(37)	(0.12)	(0.05)
Switch later	(Omitted)	(Omitted)	(Omitted)	(Omitted)
Demographics				
Age	−0.01	−5	−0.01	0.01
	(0.003)	(1.8)	(0.006)	(0.002)
Female	0.18	94	−0.08	−0.07
	(0.08)	(39)	(0.13)	(0.06)
Job Tenure	0.002	−2.3	0.002	0.003
	(0.003)	(1.6)	(0.004)	(0.002)
Income	0.003	6	0.001	−0.0003
	(0.002)	(0.9)	(0.003)	(0.001)
Health risk score				
1st quartile (< 1.119)	(Omitted)	(Omitted)	(Omitted)	(Omitted)
2nd quartile (1.119 to 1.863)	0.91	305	0.13	−0.41
	(0.07)	(59)	(0.29)	(0.06)
3rd quartile (1.863 to 2.834)	1.48	242	1.79	−0.66
	(0.08)	(81)	(0.27)	(0.06)
4th quartile (> 2.834)	2.05	−416	3.38	−0.89
	(0.09)	(120)	(0.22)	(0.07)
2004 Time dummy	−0.12	—	—	—
	(0.02)			
Variance-covariance matrix	$\overline{\mu_\lambda}$	$\ln(\omega)$	$\ln(\psi)$	
$\overline{\mu_\lambda}$	0.20	−0.03	−0.12	
	(0.03)	(0.04)	(0.02)	
$\ln(\omega)$	—	0.98	−0.01	
		(0.08)	(0.03)	
$\ln(\psi)$	—	—	0.25	
			(0.02)	
Additional parameters				
σ_μ	0.33			
	(0.03)			
σ_κ	290			
	(12)			
γ_1	0.04			
	(0.004)			
γ_2	15			
	(1.2)			

Notes: The table presents our baseline parameter estimates based on our baseline sample of 7,570 employees. As described in the text, the estimates are based on a Gibbs sampler; the table reports the posterior mean and the posterior standard deviations in parentheses. All time-varying demographics are set to their mean over the two years, except for health risk score, which we allow to be a time-varying shifter of μ_λ. "Treatment group" dummies refer to indicator variables based on the year your union's benefits were switched to the new benefits. Higher risk scores correspond to worse predicted health.

(a) Table 7A: Parameter estimates.

Table 7B—Implied Quantities

	$E(\lambda)$	ω	ψ
Unconditional statistics			
Average	4,340	1,330	0.0019
	(200)	(59)	(0.00002)
SD	5,130	3,190	0.0020
	(343)	(320)	(0.00007)
Unconditional correlations			
$E(\lambda)$	1.00	0.24	−0.36
		(0.03)	(0.01)
ω	—	1.00	−0.15
			(0.01)
ψ	—	—	1.00
Marginal effects on $E(\lambda)$			
Coverage tier			
Single	(Omitted)		
Family	360		
	(75)		
Emp+Spouse	700		
	(85)		
Emp+Children	300		
	(69)		
Treatment group			
Switch 2004	260		
	(71)		
Switch 2005	−320		
	(67)		
Switch 2006	640		
	(70)		
Switch later	(Omitted)		
Demographics			
Age	−42		
	(3.6)		
Female	720		
	(71)		
Job tenure	5.9		
	(3.1)		
Income	13		
	(2)		
Health risk score			
1st quartile (< 1.119)	(Omitted)		
2nd quartile (1.119 to 1.863)	1,600		
	(96)		
3rd quartile (1.863 to 2.834)	4,000		
	(200)		
4th quartile (> 2.834)	8,500		
	(370)		
2004 Time dummy	−590		
	(29)		

Notes: The table reports some implied quantities of interest that are derived from the estimated parameters in Table 7A. Posterior standard deviations are in parentheses.

(b) Table 7B: Implied quantities.

6.6 Table 8 and Figure 2: Model fit

Read:

- Table 8 compares actual and predicted plan-choice probabilities.
- The model fits choices under the old options especially well and gives a reasonable fit for the new options.
- Figure 2 compares actual and simulated spending distributions under the old and new options.
- The model captures the broad skewness of medical spending and the difference between the old and new option sets.

Economic message: The model fit matters because the selection-on-moral-hazard conclusions come from a structural model. The fit is not perfect, but the model captures the key choice and spending patterns well enough to support counterfactual exercises.

Table 8—Model Fit: Choice Probabilities (*Percent*)

Plan	Data	Model
Original options ($N = 6{,}896$)		
Option 1	1.2	2.0
Option 2	58	57
Option 3	41	41
New options ($N = 674$)		
Option 1	5.9	5.0
Option 2	0.5	5.0
Option 3	1.9	1.0
Option 4	27	11
Option 5	65	76

Notes: The table reports the actual and predicted choice probabilities of each plan. Plans are numbered from lowest to highest coverage. For plan details see Table 2.

(a) Table 8: Model fit for choice probabilities.

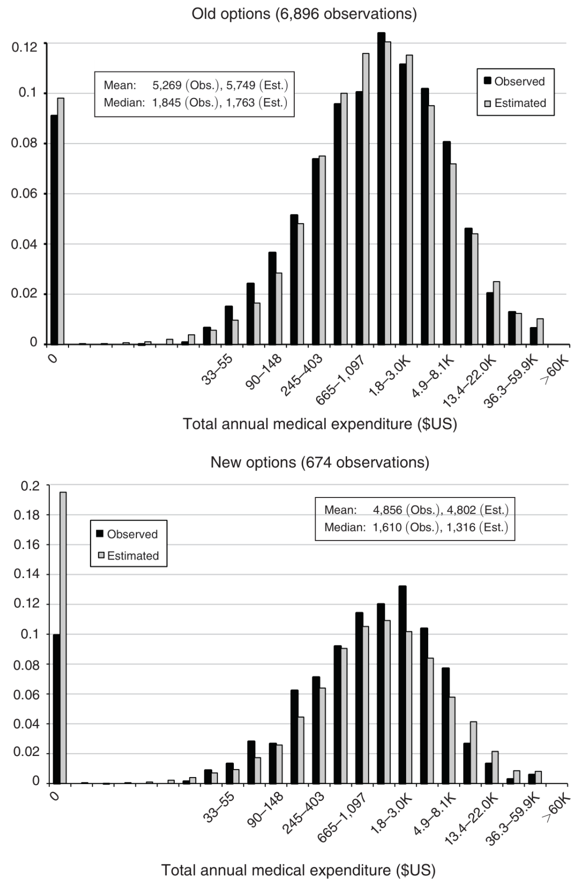

Figure 2. Model Fit: Medical Spending Distributions

Notes: The figure presents the distribution of total annual medical expenditure, in the data and in model simulations based on the estimated parameters. The graph uses a log scale, such that the second bin covers expenditure lower than exp(0.5), the next covers expenditures between exp(0.5) and exp(1), and so on; the *x*-axis labels show the corresponding dollar amounts of selected bins. The top panel compares spending of individuals who faced the original options, and the bottom panel compares the spending distribution of individuals who faced the new options.

(b) Figure 2: Model fit for medical spending distributions.

6.7 Table 9 and Figure 3: Heterogeneity and selection on moral hazard

Read:

- Table 9 quantifies spending changes from moving people across contracts.
- Moving from the no-deductible plan to the high-deductible plan reduces average spending by about $348, but the distribution is highly skewed.
- Moving from full insurance to no insurance reduces average spending by about $1,273.
- Figure 3 compares selection on moral hazard, expected health risk, and risk aversion.
- Higher moral hazard types are less likely to choose the high-deductible plan.
- The selection on moral hazard is about as important as selection on expected health risk, and more important than selection on risk aversion in this counterfactual choice.

Economic message: This is the paper's core result. People do not only select insurance based on how sick they expect to be; they also select based on how much they expect their spending to respond to insurance.

TABLE 9—SPENDING IMPLICATIONS OF MORAL HAZARD ESTIMATES

	Mean	SD	10th	25th	50th	75th	90th
Spending difference as we move from no to high-deductible plan	348	749	0	0	48	316	1,028
Spending difference as we move from full to no insurance	1,273	3,181	0	86	310	1,126	3,236

Notes: The table reports the implied spending implications if we move different employees across plans. For each employee, we use the model estimates to compute his decline in expected annual expenditure as we change his insurance plan. In the top row, we move each employee from the highest coverage option under the new benefit options (option 5) to the lowest coverage option under the new benefit options (option 1); roughly speaking, this entails moving from a plan with no deductible to a plan with a high deductible; see Table 2 for more details. In the bottom row, we move each employee from full to no insurance. The table then summarizes the cross-sectional distribution of the spending effects. The estimates are primarily driven by the estimated distribution of ω, but they take into account the truncation of spending at zero by integrating over the conditional (on ω) distribution of λ.

(a) Table 9: Spending implications of moral hazard estimates.

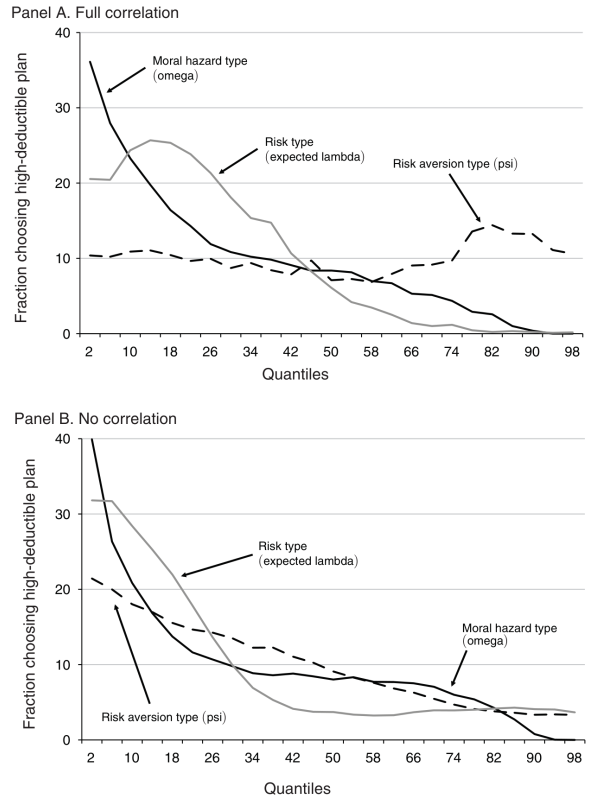

FIGURE 3. SELECTION ON MORAL HAZARD RELATIVE TO OTHER SOURCES OF SELECTION

Notes: The figure illustrates the relative importance of the three different sources of selection that we model. We consider an individual's choice between two available options: the no-deductible and high-deductible plans among the new set of options (see Table 2, options 5 and 1, respectively). We assume the observed (averaged within each coverage tier) premiums for these two options. Each point in the figure indicates the fraction of individuals choosing the high-deductible (i.e., low-coverage) option relative to the no-deductible (high-coverage) option. We consider three sources of selection: $E(\lambda)$ (risk), ω (moral hazard), and ψ (risk aversion). For each of them, we compute the fraction choosing the high deductible at different quantiles of the distribution. In the top panel, we take into account the correlation between each component and the others, while in the bottom panel we repeat the same exercise but draw the other components of the model randomly from their marginal distribution (that is, assuming no correlation).

(b) Figure 3: Selection on moral hazard relative to other sources of selection.

6.8 Figure 4 and Table 10: Spending and welfare implications

Read:

- Figure 4 shows the spending reduction from people who endogenously choose the high-deductible plan as the relative price of generous coverage changes.
- When only a small share of employees chooses the high-deductible plan, the spending reduction among those choosers is much smaller than the average effect for everyone.
- This is because the early adopters of high deductibles are low-moral-hazard types.

- Table 10 reports welfare counterfactuals for screening and monitoring.
- Perfect screening eliminates adverse selection by pricing on health risk and moral hazard.
- Perfect monitoring eliminates moral hazard by reimbursing only spending related to underlying health needs.
- The welfare gain from eliminating moral hazard is much larger than the welfare gain from eliminating adverse selection, but selection on moral hazard is still a nontrivial part of the adverse-selection cost.

Economic message: Ignoring selection on moral hazard can overstate the spending effect of high-deductible plan take-up. Contract design that changes moral hazard can also change selection because the slope of spending is part of what people privately know about themselves.

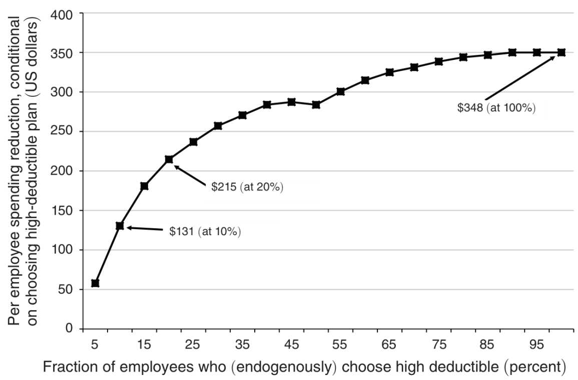

FIGURE 4. SPENDING IMPLICATIONS OF SELECTION ON MORAL HAZARD

Notes: The figure illustrates the potential spending implications arising from selection on moral hazard. To construct the figure, we use an exercise similar to the one used for Figure 3. For each individual, we use the model estimates to compute his decline in expected annual expenditure as we move him from the highest coverage (no deductible) to the lowest coverage (high deductible) in the new benefits options (see Table 2, options 5 and 1, respectively). We then vary the relative price of the highest coverage, allowing employees to choose endogenously between the two options, and report the per-employee expected decrease in spending for the group of individuals who choose the lowest coverage at each price. Without selection on moral hazard, the curve would have been flat. Selection on moral hazard implies that those with the lowest moral hazard effects of insurance are those who have the lowest willingness to pay for incremental coverage and are therefore the first (as the price of coverage increases) to switch from higher to lower coverage. Ceteris paribus, therefore, selection on moral hazard generates an upward-sloping curve; this can be offset through the correlation between moral hazard and other components of demand (such as risk aversion or health risk).

(a) Figure 4: Spending implications of selection on moral hazard.

TABLE 10—SPENDING AND WELFARE EFFECTS OF ASYMMETRIC INFORMATION

	Average equilibrium (incremental) premium	No-deductible plan share	Expected spending per employee	Total welfare per employee
(1) "Status quo": no screening or monitoring	1,568	0.90	5,318	Normalized to 0
(2) "Perfect screening": premiums depend on F(λ) and ω	1,491	0.91	5,248	52
(3) "Imperfect screening": premiums depend on ω (but not on F(λ))	1,523	0.88	5,265	34
(4) "Perfect monitoring": contracts reimburse only "λ-related" spending	1,139	0.94	4,185	490
(5) "Imperfect monitoring": perfect monitoring assumed for choice (but not for utilization)	1,139	0.94	5,327	25

Notes: The table reports the spending and welfare effects from a set of counterfactual contracts described in the text. All exercises are applied to a setting in which the only two options available are the no-deductible plan and the high-deductible plan under the new benefit options (i.e., option 5 and option 1, respectively; see Table 2). Equilibrium premiums are computed as the incremental (relative) premium for the no-deductible plan that equals the expected incremental costs associated with providing the no-deductible plan to those who choose it. The no-deductible plan share is calculated based on the choice probabilities as a function of equilibrium premiums. Expected spending and total welfare are computed based on these choices. Row 1 assumes the "status quo" asymmetric information contracts, with a "uniform" price that varies only by coverage tier. Row 2 assumes "perfect screening," so that contracts are priced based on ω_i and all components of $F_i(\lambda)$, and adverse selection is eliminated. Row 3 assumes "imperfect screening," in which contracts are priced based only on ω_i. Row 4 assumes "perfect monitoring" so that moral hazard is eliminated. Specifically, we assume the insurance provider can counterfactually observe (and not reimburse) spending that is associated with moral hazard; spending associated with health—realization of λ—is reimbursed according to the observed contracts. Row 5 assumes "imperfect monitoring" in which, ex ante, individuals choose contracts under the assumption that there will be perfect monitoring (i.e., spending associated with moral hazard will not be reimbursed), but ex post (after they choose their contract but before they make their spending decision) the contracts are changed to be the standard contracts that reimburse all medical spending regardless of its origin.

(b) Table 10: Spending and welfare effects of asymmetric information.

7 Short summary

- The paper studies health insurance choices and medical spending at Alcoa.
- The central concept is selection on moral hazard: people may choose coverage based on their expected behavioral response to insurance.
- The model separates expected health risk, risk aversion, and moral hazard.
- A staggered transition to higher-cost-sharing PPO options identifies average moral hazard.
- The structural model estimates large heterogeneity in moral hazard.
- Higher moral hazard individuals are more likely to demand generous coverage.

- In the key high-deductible versus no-deductible counterfactual, selection on moral hazard is roughly as important as selection on expected health risk.
- Ignoring selection on moral hazard can overstate the spending reduction from introducing high-deductible plans.
- Welfare exercises show that selection on moral hazard matters for adverse selection, but the traditional moral hazard distortion is larger in this setting.

8 Conclusion

8.1 Core conclusion

Einav, Finkelstein, Ryan, Schrimpf, and Cullen show that adverse selection in health insurance can operate not only through the expected level of health spending but also through the behavioral slope of spending with respect to coverage. Individuals who would increase spending more when insured are more likely to choose more generous coverage.

8.2 Economic intuition

A person who expects insurance to change medical use by a lot receives more value from generous coverage than a person whose utilization would barely change. This makes high-ω individuals more willing to pay for generous plans. Insurers then face a form of adverse selection based on the cost created by the insurance contract itself.

8.3 Incremental contribution revisited

The paper adds a new dimension to empirical insurance-market analysis. Instead of treating adverse selection and moral hazard as separate objects, it shows how they interact. The conceptual innovation is to decompose selection into selection on levels and selection on slopes. The empirical innovation is to quantify that decomposition using plan choices, spending, and a staggered cost-sharing change.

8.4 Implications for health insurance design

The findings matter for high-deductible plans and for policy forecasts. Randomized or quasi-random estimates of average moral hazard may not predict what happens when people voluntarily choose high deductibles, because the people who choose them are selected. The findings also imply that monitoring tools, such as differential cost sharing across services with different behavioral responsiveness, may reduce both moral hazard and adverse selection.

8.5 Limitations

The results come from one employer and one institutional setting. The structural model depends on assumptions about correct beliefs, functional forms, and latent heterogeneity. Spending responses may include both low-value and high-value care, so welfare conclusions require caution. The results are therefore best read as evidence that selection on moral hazard is real and quantitatively meaningful in this context, not as a universal estimate for all health insurance markets.

8.6 Takeaway

The lasting takeaway is that people privately know not only something about their health risk, but also something about how they will behave when insured. This behavioral private information changes how insurance choices should be interpreted and how high-cost-sharing policies should be evaluated.